1. Triplets

Given an array of *n* distinct integers, $d = [d[0], d[1], ..., d[n-1]]$, and an integer threshold, *t*, determine the number of *(a, b, c)* index triplets that satisfy both of the following conditions.

- $d[a] < d[b] < d[c]$
- $d[a] + d[b] + d[c] \le t$

Example

$d = [1,2,3,4,5]$

$t = 8$

The following 4 triplets satisfy the constraints:

```
(1,2,3) → 1 + 2 + 3 = 6 ≤ 8
(1,2,4) → 1 + 2 + 4 = 7 ≤ 8
(1,2,5) → 1 + 2 + 5 = 8 ≤ 8
(1,3,4) → 1 + 3 + 4 = 8 ≤ 8
```

Function Description

Complete the function *triplets* in the editor below.

triplets has the following parameter(s):

int t: the threshold

int d[n]: an array of integers

Returns:

long: a long integer that denotes the number of *(a, b, c)* triplets that sastisfy the given conditions

Constraints

- $1 \le n \le 10^4$

Autocomplete Ready

```
> #include <assert.h>…

// Complete the tripl
long triplets(long t,


}

> int main()…
```

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1

Constraints

- $1 \leq n \leq 10^4$
- $0 \leq d[i] < 10^9$
- $0 < t < 3 \times 10^9$

▼Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The first line contains an integer *t*, the given integer threshold.
The second line contains an integer *n*, the size of the array *d*.
Each of the next *n* lines contains an integer *d[i]* where $0 \leq i < n$.

▼Sample Case 0

Sample Input

```
STDIN       Function
-----       -----
8       →   integer threshold t = 8
5       →   size of d[] n = 5
1       →   d = [1, 2, 3, 4, 6]
2
3
4
6
```

Sample Output

```
3
```

Explanation

Language C++14

Autocomplete Ready

```
#include <bits/stdc

// Complete the tr
long triplets(long


}

int main() ⋯
```

The first line contains an integer t, the given integer threshold.
The second line contains an integer n, the size of the array d.
Each of the next n lines contains an integer $d[i]$ where $0 \le i < n$.

▼ Sample Case 0

Sample Input

```
STDIN       Function
-----       -----
8       →   integer threshold t = 8
5       →   size of d[] n = 5
1       →   d = [1, 2, 3, 4, 6]
2
3
4
6
```

Sample Output

```
3
```

Explanation

Given $t = 8$ and $d = [1, 2, 3, 4, 6]$, the following triplets satisfy the conditions:

1. $(0, 1, 2) \rightarrow 1 + 2 + 3 \le 8$
2. $(0, 1, 3) \rightarrow 1 + 2 + 4 \le 8$
3. $(0, 2, 3) \rightarrow 1 + 3 + 4 \le 8$

▶ Sample Case 1

```
// Complete
long triple

}

int main()
```

Test Results

Explanation

Given $t = 8$ and $d = [1, 2, 3, 4, 6]$, the following triplets satisfy the conditions:

1. $(0, 1, 2) \rightarrow 1 + 2 + 3 \leq 8$
2. $(0, 1, 3) \rightarrow 1 + 2 + 4 \leq 8$
3. $(0, 2, 3) \rightarrow 1 + 3 + 4 \leq 8$

▼ Sample Case 1

Sample Input

```
STDIN       Function
-----       -----
7       →   integer threshold t = 7
4       →   size of d[] n = 4
3       →   d = [3, 1, 2, 4]
1
2
4
```

Sample Output

```
2
```

Explanation

Given $t = 7$ and $d = [3, 1, 2, 4]$, the following triplets satisfy the conditions:

1. $(1, 2, 0) \rightarrow 1 + 2 + 3 \leq 7$
2. $(1, 2, 3) \rightarrow 1 + 2 + 4 \leq 7$

```
long tripl

}

> int main(
```

Test Results

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9	Client 1 Left and Client 4 Joined	4
10		4
11	Client 4 Left	

ALL

The maximum number of concurrent interactions is 3 which happens first at the 6^{th} second. Return 6.

Function Description

Complete the function *getMaxTrafficTime* in the editor below.

getMaxTrafficTime has the following parameters:

int start[n]: interaction start times

int end[n]: interaction end times

Returns

int: the earliest time of maximum concurrent interactions

Constraints

- $1 \le n \le 10^5$
- $1 \le start[i] \le end[i] \le 10^9$

▼Input Format for Custom Testing

The first line contains an integer *n*, the number of clients.

Each of the next *n* lines contains an integer *start[i]*.

The next line contains the same integer *n*, the number of clients.

Each of the next *n* lines contains an integer *end[i]*.

▼Sample Case 0

Sample Input 0

```
STDIN       FUNCTION
-----       --------
```

Language C++14

Autocomplete Ready

```
> #include <bits/stdc++
/*
 * Complete the 'getl
 *
 * The function is
 * The function acc
 *  1. INTEGER_ARRA
 *  2. INTEGER_ARR
 */

int getMaxTraffic
{

}

> int main()…
```

Test Results

left

ALL

1

2

3

```
-----      --------
5     →    start[] size n = 5
2     →    start[] = [2, 3, 7, 4, 7]
3
7
4
7
5     →    end[] size n = 5
4     →    end[] = [4, 5, 8, 7, 10]
5
8
7
10
```

Sample Output 0

```
4
```

Explanation

At time $t = 4$, the first, second, and fourth clients are connected to the server. At time $t = 7$, the last three clients are connected.

The earliest time when 3 are connected is $t = 4$.

▾ Sample Case 1

Sample Input 1

```
STDIN      FUNCTION
-----      --------
3     →    start[] size n = 3
1     →    start[] = [1, 1, 1]
1
1
3     →    end[] size n = 3
5     →    end[] = [5, 5, 5]
5
5
```

Sample Output 1

Language C++14

Autocomplete Ready

```
#include <bits/stdc++.h>

/*
 * Complete the 'getMa
 *
 * The function is exp
 * The function accep
 *  1. INTEGER_ARRAY
 *  2. INTEGER_ARRAY
 */

int getMaxTrafficTi
{

}

int main() ⋯
```

Test Results

C

ALL

1

2

3

```
5
8
7
10
```

Sample Output 0

```
4
```

Explanation

At time $t = 4$, the first, second, and fourth clients are connected to the server. At time $t = 7$, the last three clients are connected.

The earliest time when 3 are connected is $t = 4$.

▼Sample Case 1

Sample Input 1

```
STDIN          FUNCTION
-----          --------
3       →      start[] size n = 3
1       →      start[] = [1, 1, 1]
1
1
3       →      end[] size n = 3
5       →      end[] = [5, 5, 5]
5
5
```

Sample Output 1

```
1
```

Explanation

All three clients are connected to the server at $t = 1$.

```
#include <bits/stdc+
/*
 * Complete the 'ge
 *
 * The function is
 * The function ac
 *  1. INTEGER_ARR
 *  2. INTEGER_AR
 */

int getMaxTraffi
{

}

int main()…
```

Test Results

3. SQL: Calendar Application Events Report 2

You are working on a calendar application and need to get a list of the five earliest events with all the guests participating in those events.

The result should have the following columns: *dt* / *title* / *guests*.

- *dt* - event timestamp
- *title* - event name
- *guests* - list of guest records participating in the event:
    - Record format is ## <##>, where the placeholders are ## in the order they appear:
        1. Full name of the guest
        2. Guest email address
    - Records are separated by a comma
    - Records are sorted in ascending order

The result should be sorted in ascending order by *dt*.

The result should be limited to the first five records.

Note:

- Only events whose guests are not on vacation should be included in the report.

▾Schema

events		
name	**type**	**description**
id	SMALLINT	Event ID
dt	VARCHAR(19)	Timestamp

Language

MySQL

```
/*
Enter your query below.
Please append a semicolon '
*/
```

Test Results